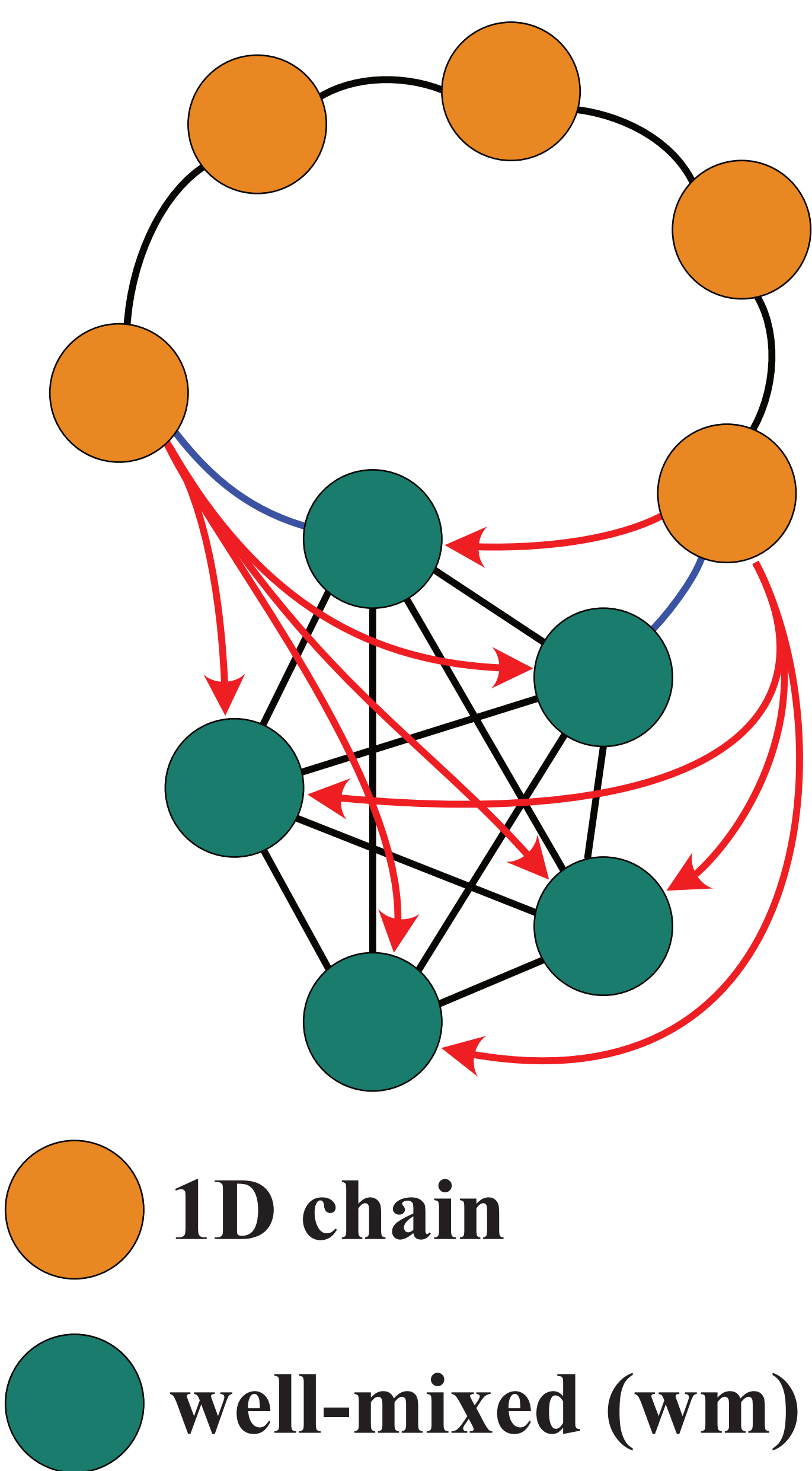


Fitness landscape topology determines evolutionary hotspot location in spatially heterogeneous population structures

Clifford Bohm, Grant Gordon, Austin J. Ferguson, Alexander Lalejini



How does spatial structure effect the evolutionary process?

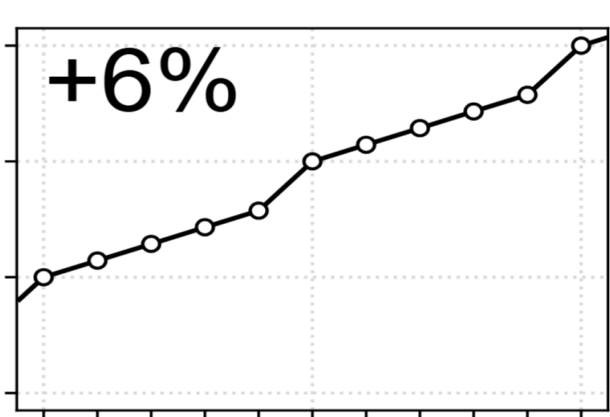
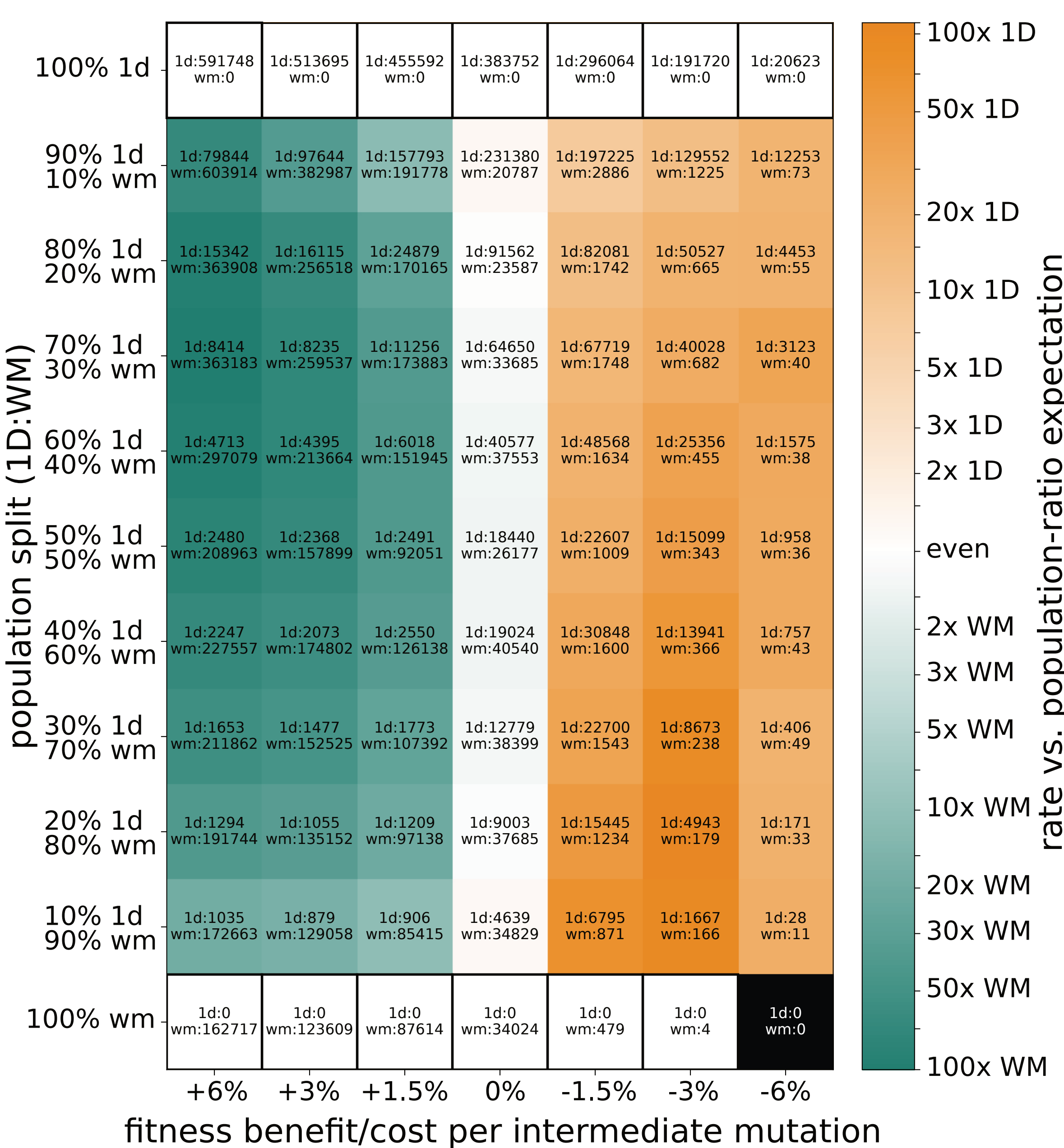
- Densely connected regions (wm) strengthen selection and speed the spread of beneficial mutations.
- Sparsely connected regions (1D) reduce competition and can enhance discovery on rugged (i.e., deceptive) landscapes.

What happens if both population structures coexist?

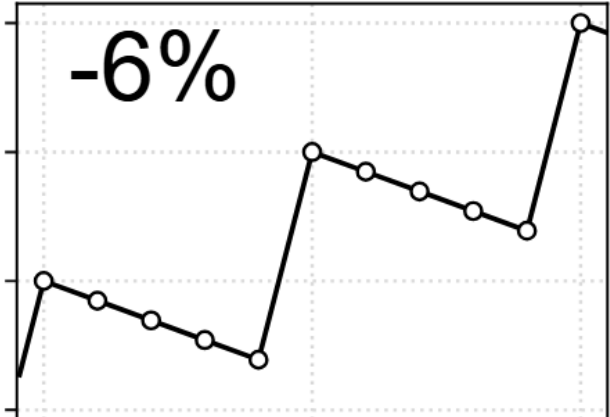
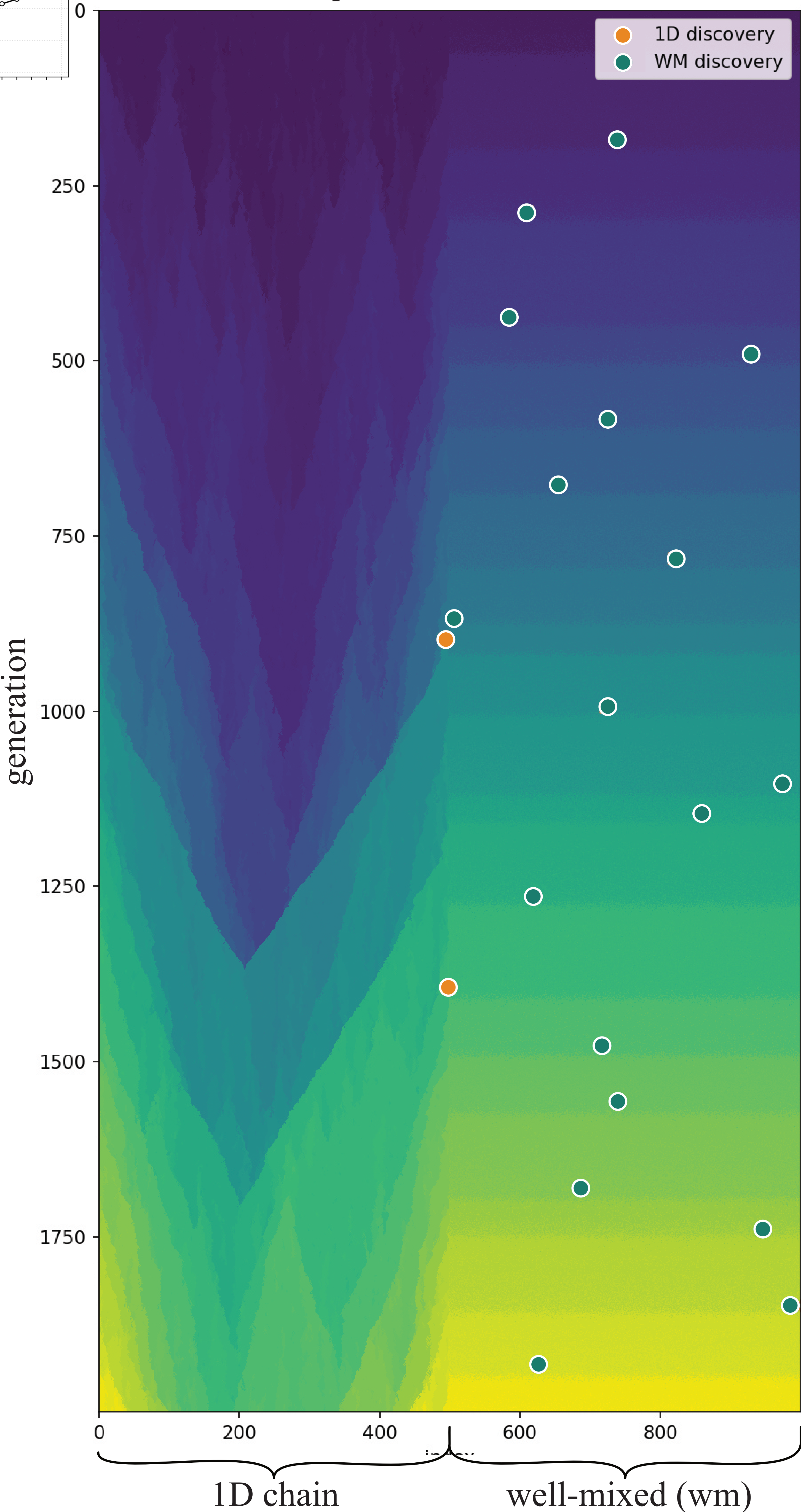
- Populations of 1000 agents were evolved on graphs containing different sized linear-chain (1D) and well-mixed (wm) regions across smooth, neutral, and rugged fitness landscapes.

Location of evolutionary discoveries depends on local population topology.

- On smooth fitness landscapes, discoveries occurred disproportionately in the well-mixed region (wm).
- On rugged landscapes, discoveries occurred disproportionately in the 1D chain region (1D).
- On neutral landscapes discovery location was roughly proportional to size of each region.



Smooth Landscape



Rugged Landscape

